

Asif Ahmad

Incoming PhD Researcher in Computer Science @ University of Leicester

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Education

- **University of Leicester, United Kingdom** from Sep. 2026
Doctor of Philosophy in Computer Science
Field of study: **Agentic Artificial Intelligence for Federated Network Digital Twins**
Funding: EPSRC Industrial Doctoral Landscape Award (IDLA) studentship in partnership with BT
- **University of Engineering & Technology Peshawar, Pakistan** Mar. 2018 – Apr. 2020
Master of Science (18-Year) in Applied Mathematics Supervisor: Dr. Noor Badshah, Ph.D (UK)
Thesis Title: **Neural Network for Segmentation of Medical Images**
- **University of Peshawar (UOP), Pakistan** Nov. 2014 – Mar. 2017
Master of Science (16-Year) in Mathematics

Work Experience

- **Lecturer**, GIK Institute, Pakistan Feb. 2022 – Aug. 2026
Taught undergraduate mathematics and machine learning in the Faculty of Engineering and Basic Sciences. Principal Investigator on a funded R&D project for NADRA, delivered in November 2025. Co-supervised an MS thesis on physics-informed neural networks and Lie symmetry. Undergraduate courses taught:
 - Machine Learning (ES 442) Fall 2025
 - Calculus I (MT 101) Fall 2023, 2024, 2025
 - Advanced Linear Algebra (ES 205) Spring 2025
 - Complex Variables & Transforms (MT 203) F. 2024, S. 2026
 - Probability & Statistics (ES 111) Spring 2024
 - Differential Equations & Linear Algebra-I (MT 102/202) F. 2022, S. 2023, 2026
 - Engineering Statistics (ES 202) Spring 2022
- **Co-founder and ML Team Lead**, Neural Internet, USA Sep. 2022 – May 2025
 - Led a Machine Learning team focused on developing decentralized applications leveraging various subnets within the Bittensor network.
 - Provided technical guidance and collaborated with the development team on the Compute Subnet (SN-27), contributing technical insights and suggesting improvements.
 - Contributed to the maintenance and operational stability of network validators on the Bittensor network.
 - Collaborated with the Bittensor community to advance decentralized AI capabilities.
- **Mathematics Lecturer**, CECOS University, Pakistan Mar. 2020 – Jan. 2022
- **Mathematics Lecturer (Visiting)**, CECOS University, Pakistan Oct. 2019 – Mar. 2020
- **Research Assistant**, UET Peshawar, Pakistan Feb. 2019 – Feb. 2020
- **Mathematics Lecturer**, Govt. Degree College Chitral, Pakistan Aug. 2017 – Aug. 2018

Research Experience

AI-based Service Request Analysis and Rate Limiting for Fair and Efficient Use of NADRA Resources *PI, Completed*

Led R&D project RP #N-IS1-25-0006 for the National Database and Registration Authority (NADRA). Built an unsupervised anomaly detection model over engineered behavioural features from roughly 671,000 service requests, and designed a behaviour-driven rate-limiting framework that maps continuous anomaly scores to graduated risk controls. The retrainable framework, trained models, and validation pipeline were delivered to NADRA in November 2025. A research manuscript based on the project was prepared and shared with NADRA in July 2026.

Careem Captain Fraud Detection *Co-PI*

In this project with Careem, a well-known ride-hailing services provider in the Middle East, we explored data processing techniques and machine learning algorithms to automate Captain (driver) fraud detection.

Solving Partial Differential Equations Using PINN *Co-supervised*

Developed and trained a Physics-Informed Neural Network (PINN) for solving PDEs with a graduate student at GIK Institute. Enhanced PINN performance through Lie symmetry groups, and solved the Newell–Whitehead–Segel, Allen–Cahn, and Burgers’ equations.

Neural Network for Detection of Hidden Explosives and Narcotics Using Multi-pulse NQR

Developed a deep neural network-based approach for the rapid detection of hidden explosives and narcotics using Multipulse Nuclear Quadrupole Resonance (NQR).

Neural Network for Segmentation of Medical Images *Masters’ Thesis Work*

Developed and trained a deep neural network for the segmentation of medical images during masters’ degree research. Also collaborated with other graduate students on integrating Active Contour Models into neural networks for medical image processing.

Skin Cancer Detection with Deep Neural Networks and Vision Transformers *Book chapter*

Used transfer learning to fine-tune convolutional deep neural networks and vision transformers for classification of cancerous skin images.

Publications

Book Chapters

- Muniba Ashfaq, **Asif Ahmad**, *Skin cancer classification with convolutional deep neural networks and vision transformers using transfer learning*, Springer Nature, 2024.

Journal Articles

- Noor Badshah, **Asif Ahmad**, *ResBCU-Net: Deep learning approach for segmentation of skin images*, Biomedical Signal Processing and Control, Elsevier, 2022.
- **Asif Ahmad**, Noor Badshah, Mahmood Hassan, Kambogh Well, *A Modified Memory-Efficient U-Net for Segmentation of Polyps*, International Journal of Engineering Works, 2021.

Conference Papers

- Mehreen Fida, Noor Badshah, **Asif Ahmad**, Abdul Hafeez, Arif Ullah, Yalin Zheng, *Neural Network Integrated with Active Contour Model for Segmentation of Medical Images*, Ahi Evran International Conference on Scientific Research, 30 November – 1–2 December, 2021.

- Arif Ullah, Noor Badshah, **Asif Ahmad**, Abdul Hafeez, Mehreen Fida, *Deep Learning Approach for Segmentation of Medical Images*, Ahi Evran International Conference on Scientific Research, 30 November – 1–2 December, 2021.

Preprints

- **Asif Ahmad**, Naveed R. Butt, Babar Zaman, *Deep Neural Network Based Accurate and Fast Detection of Hidden Explosives and Narcotics Using Multi-Pulse Nuclear Quadrupole Resonance*, Available at SSRN 4948911.
- Ali Haider Shah, Naveed Razzaq Butt, **Asif Ahmad**, Muhammad Omar bin Saeed, *Enhancing PINN Performance Through Lie Symmetry Group*, arXiv.org e-Print archive, 2025.
- Ali Haider Shah, Naveed Razzaq Butt, **Asif Ahmad**, Muhammad Omar bin Saeed, *Solving Newell–Whitehead–Segel and Allen–Cahn Equations Employing Physics-Informed Neural Networks: A Comparative Analysis with Spline Methods*, arXiv.org e-Print archive, 2025.

Supervision of Graduate Students

- Co-supervised MS thesis titled *Solving PDEs Using Improved Physics-Informed Neural Networks (PINNs) and Lie Symmetry* of Ali Haider (Reg. # ES-2264), at GIK Institute of Engineering Sciences & Technology Topi, Pakistan.

Conferences/Workshops

- 5-Day Training Workshop on **Teaching Statistics & ML using Python** *Instructor* Jan. 2023
- 1st Intl. Conference on Science, Engineering & Technology (**ICSET-21**) *Presenter* Mar. 2021
- National Conf. on Mathematical Sciences in Engrg. Applications (**NCMSEA**) *Organizer* Apr. 2018

Honours and Awards

- Fully funded EPSRC Industrial Doctoral Landscape Award (IDLA) studentship in partnership with BT, University of Leicester, 2026.
- MS Studentship at National Center of Big Data & Cloud Computing (NCBC), UET, 2019.

Computer Skills

Programming: Python, MATLAB

Paper Writing: L^AT_EX, MS Word